

Classifying Triangles & Triangle Angle Theorems

Geometry Worksheet · Grade 7–9

Name: _____

Date: _____

Learning Objectives

- Classify triangles by their angles (acute, right, obtuse, equiangular) and by their sides (scalene, isosceles, equilateral)
- Identify and apply the Triangle Angle Sum Theorem to find missing angle measures
- Use geometric notation correctly to name sides, angles, and vertices of a triangle

Problems

1. A triangle has three angles each measuring 60° . What type of triangle is this classified as according to its angles?

$$60^\circ + 60^\circ + 60^\circ = 180^\circ$$

2. In triangle PQR, which angle is opposite side PQ?

$\triangle PQR$

3. A triangle has one angle measuring 110° . Classify this triangle by its angles.

$$110^\circ > 90^\circ$$

4. In triangle ABC, side AC is included between which two angles?

$\triangle ABC$

5. All three angles of a triangle are less than 90° . What type of triangle is it?

$$\angle A < 90^\circ, \angle B < 90^\circ, \angle C < 90^\circ$$

6. Using the Triangle Angle Sum Theorem, find the missing angle x in a triangle where the other two angles measure 45° and 85° .



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$$45^\circ + 85^\circ + x = 180^\circ$$

7. In triangle EBA, list all three vertices and one possible correct way to name the triangle using those vertices.

$\triangle EBA$

8. A triangle has one angle measuring exactly 90° . Classify this triangle by its angles and find the sum of the remaining two angles.

$$90^\circ + \angle A + \angle B = 180^\circ$$

9. Two angles of a triangle are congruent and each measures 40° . Classify the triangle by its angles and find the third angle.

$$40^\circ + 40^\circ + x = 180^\circ$$

10. In triangle PQR, the measure of angle P is $(3x + 5)^\circ$, angle Q is $(2x - 10)^\circ$, and angle R is $(x + 15)^\circ$. Find the value of x and classify the triangle by its angles.

$$(3x + 5) + (2x - 10) + (x + 15) = 180$$



Classifying Triangles & Triangle Angle Theorems — Answer Key

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Answer Key

1. Answer: Equiangular triangle

- All three angles are equal (each 60°), so all angles are congruent.
- A triangle with all equal angles is called an equiangular triangle.

2. Answer: Angle R

- Side PQ connects vertices P and Q.
- The angle opposite a side is located at the vertex NOT on that side, which is vertex R.

3. Answer: Obtuse triangle

- An obtuse angle is any angle greater than 90° .
- Since one angle measures 110° , the triangle is classified as an obtuse triangle.

4. Answer: Angle A and Angle C

- An included side is the side that lies between two angles.
- Side AC connects vertex A and vertex C, so it is included between angle A and angle C.

5. Answer: Acute triangle

- An acute angle measures less than 90° .
- Since all three angles are less than 90° , the triangle is an acute triangle.

6. Answer: 50°

- The sum of all angles in a triangle equals 180° .
- $x = 180^\circ - 45^\circ - 85^\circ = 50^\circ$.

7. Answer: Vertices: E, B, A — e.g., $\triangle ABE$ or $\triangle BAE$

- The vertices are the three non-collinear points that form the triangle: E, B, and A.
- The triangle can be named using any order of those three letters, such as $\triangle ABE$ or $\triangle BAE$, as long as all three vertices are included.

8. Answer: Right triangle; remaining angles sum to 90°

- A triangle with one 90° angle is classified as a right triangle.
- The remaining two angles must sum to $180^\circ - 90^\circ = 90^\circ$.

9. Answer: Acute triangle; third angle = 100° — wait, $100^\circ > 90^\circ$, so Obtuse triangle; third angle = 100°

- $x = 180^\circ - 40^\circ - 40^\circ = 100^\circ$.
- Since one angle (100°) is greater than 90° , the triangle is an obtuse triangle.



10. Answer: $x = 28.33^\circ$; angles $\approx 90^\circ, 46.67^\circ, 43.33^\circ \rightarrow$ Right triangle

- Combine like terms: $6x + 10 = 180$, so $6x = 170$, giving $x = \frac{170}{6} \approx 28.33$.
 - Angle $P = 3(28.33) + 5 \approx 90^\circ$, Angle $Q = 2(28.33) - 10 \approx 46.67^\circ$, Angle $R = 28.33 + 15 \approx 43.33^\circ$; since one angle is approximately 90° , this is a right triangle.
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